# Low-tech graphite sensor

## Main features

- Low power usage (3.3V-5V)
- Low cost
- Eco-friendly
- Easy to use
- Ergonomic and portable
- Easily repairable

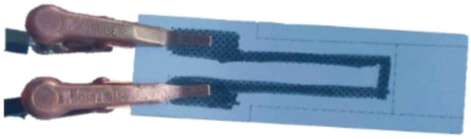

*Figure 1 : Graphite sensor*

## Applications

- Tests findings of *Pencil Drawn Strain Gauges and Chemiresistors on Paper*
- Applied using a transimpedance amplifier connected to the ADC of an Arduino Uno Card
- Pedagogical tool for students to create and implement their own PCB design

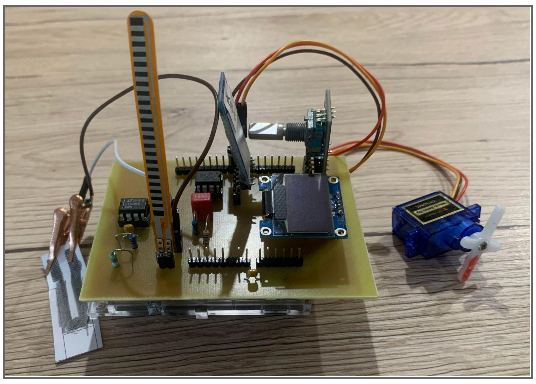

*Figure 2 : Graphite sensor connected to the PCB*

## General description

This innovative sensor, conceptualized and made by students from Engineering Physics Department of INSA Toulouse, is a tool inspired by the publication *Pencil Drawn Strain Gauges and Chemiresistors on Paper*[1].
This research paper provides a simple, cost-efficient, and highly pedagogical tool for students to master their skills in Physics, Electronics, Programing, and Sensor Design. The sensor presented in the publication is a small piece of paper coated with a graphite layer from a pencil.

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[1] by LIN Cheng-Wei, ZHAO Zhibo, KIM Jaemyung et HUANG Jiaxing, 2014. Pencil Drawn Strain Gauges and Chemiresistors on Paper. Scientific Reports. 22 janvier 2014. Vol.4, n°1, pp.3812. DOT 10.1038/srep03812.

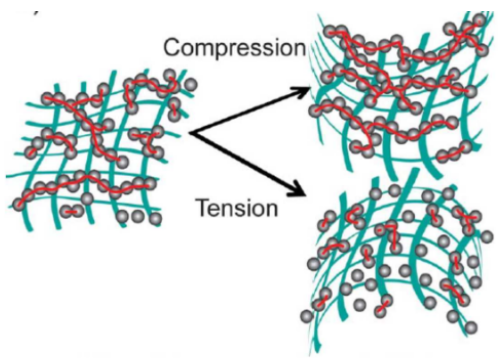


*Figure 3 : Compression & tension in a granular system*

Due to the deposited graphite on the piece of paper, the electrons are able to move freely from particle to particle due to quantum tunneling. This effect is extremely sensitive to the slightest movement of the paper. We observe that compressing or stretching the graphite will change the resistivity and the conductivity of the sensor. Thanks to this, we will be able to trace the deformation like a traditional strain gauge.

The structure of the graphite layer depends on the type of pencil used. We conducted tests with different types of pencils.

To achieve this, a transimpedance amplifier is needed to amplify the signal and then calculate the deformation. The transimpedance amplifier, the sensor and the other components are mounted on a PCB and set on an Arduino Uno card.

## Dimensional and electrical diagram

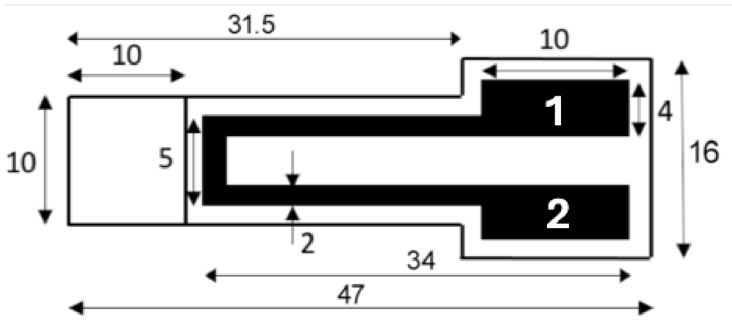


*Figure 4 : Top view - Dimensions (mm)*

Graphite layer

Thickness : 0.2mm

| Pin | Symbol | Parameter |
|---|---|---|
| 1 | Vcc | Supply voltage |
| 2 | Vout | Out sensor |

*Table 1 : Specifications of the electrical diagram*

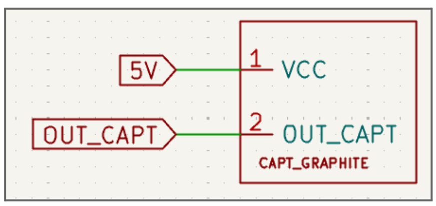


*Figure 5 : Sensor's schematic*

Note : Vcc is typically a +5V voltage. Both pins are reversible.

# Technical specifications

| Sensor | Strain sensor |
|---|---|
| Sensor Type | Passive<br>(5V power supply required) |
| Materials | Graphite<br>(4B to 2H pencils) |
| Nature of output signal | Analog |
| Nature of measurand | Voltage |
| Typical response time | < 50 ms |
| Typical use | Evaluation of a compression/tension deformation |

*Table 2 : Technical specifications of the graphite sensor*

# Absolute maximum ratings

| Parameter | Symbol | Unit | Value | | |
|---|---|---|---|---|---|
| | | | Min | Typical | Max |
| Supply voltage | Vcc | V | - | 5.0 | - |
| Temperature | T | °C | 10 | 20 | 30 |
| Air humidity | - | % | 30 | - | 60 |
| Air quality | - | %N2/O2 | - | 80/20 | - |
| Life cycle | - | - | 10 | - | 15 |
| Paper thickness | - | mm | 0.15 | - | 0.3 |
| Pencil tone[2] | - | - | 4B | - | 2H |

*Table 3 : Absolute maximum ratings of the graphite sensor*

[2] Corresponds to the US grading system.

## Electrical characteristics

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Min | Typical | Max |
| Power supply | V | 3.0 | 5.0 | 7.0 |
| HB | kΩ | 185 | 385 | 585 |
| 4B | kΩ | 25 | 37.5 | 50 |
| 5B | kΩ | 55 | 95 | 135 |

*Table 4 : Values of the sensor's resistance depending on the type of pencil*

Note : Values highly depend on the amount of graphite deposited.

## Performance characteristics

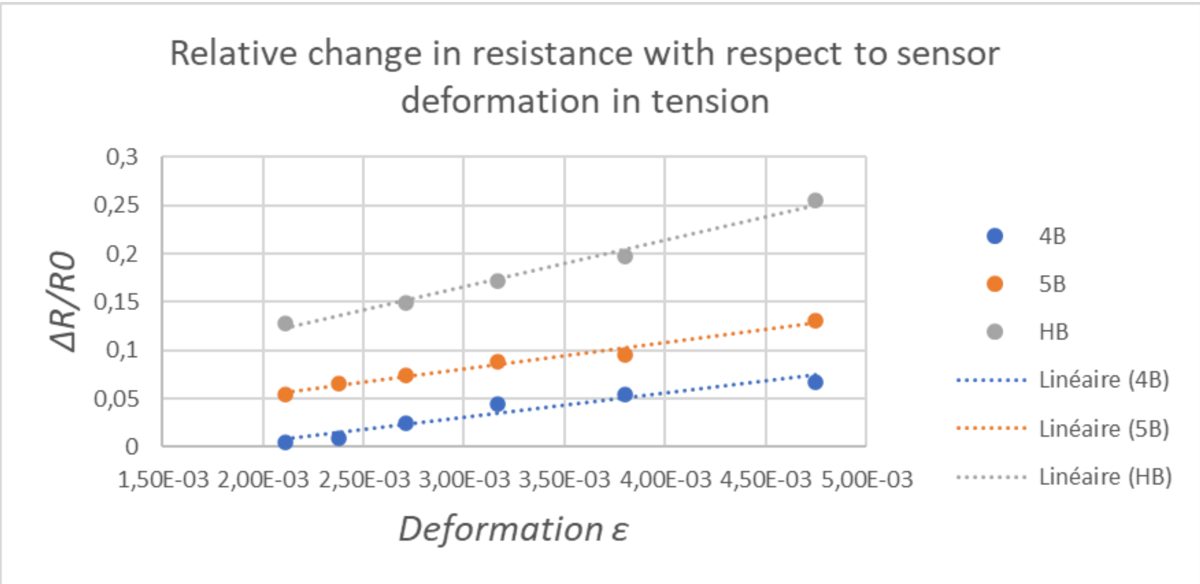


*Figure 6 : Performance characteristics for tension*

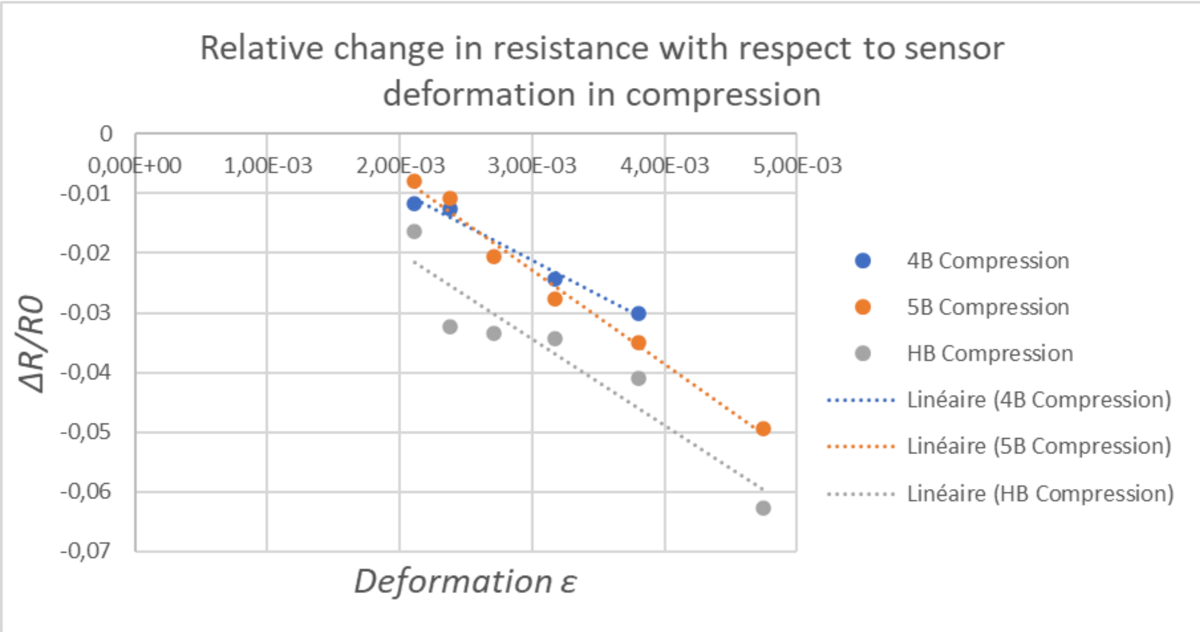


*Figure 7 : Performance characteristics for compression*

# Applications

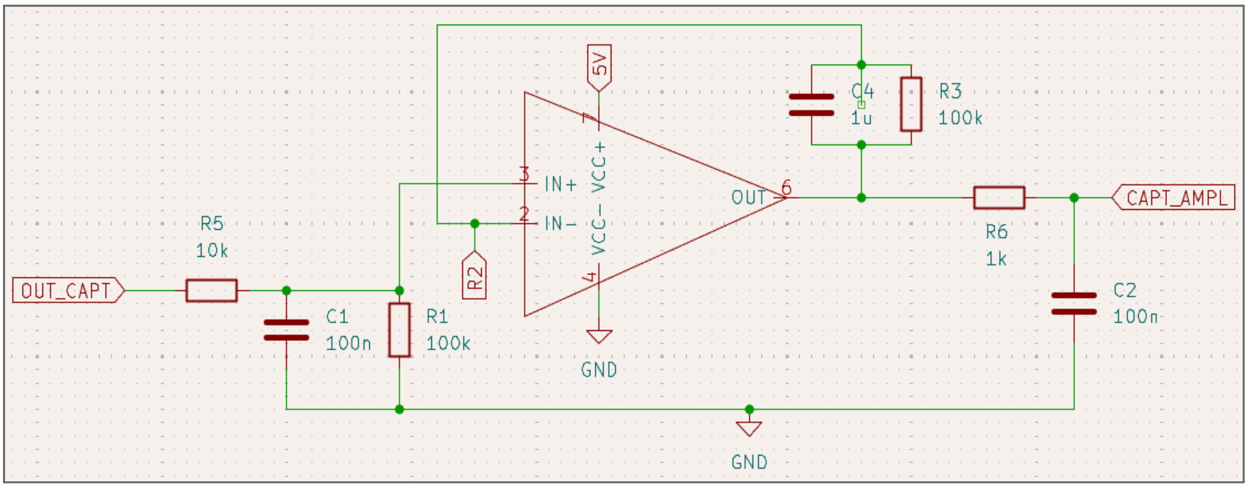


*Figure 8 : Transimpedance amplifier circuit*

The output signal of the sensor is transmitted to an Analog-to-Digital (ADC) converter on the Arduino Uno Card. We have to make sure that the signal received by the ADC is big enough to be read but does not saturate. Therefore, the sensor is connected to a transimpedance amplifier circuit.

The circuit is equipped with a low-pass and high-pass filters to cancel out the noise from the sector, the current and the amplification of the signal.

On our PCB, we have installed a digital potentiometer in lieu of the $R_2$ resistance in order to match the amplification of the circuit for each pencil tone.

We can calculate the value of the resistance of the sensor knowing the value of the final voltage, with the following formula :

$$R_{sensor} = R_1 \left(1 + \frac{R_3}{R_{variable}}\right) \frac{V_{cc}}{V_{adc}} - R_1 - R_5$$